2-D Unstructured Compressible Navier–Stokes Solver: Benchmark Report

cfd-solver v1.0

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Abstract

We present a 2-D unstructured finite-volume solver for the compressible Navier–Stokes equations, built in C++17 with MPI parallelism via METIS cell-graph partitioning and neighbor halo exchange. The solver implements Rusanov (local Lax–Friedrichs) inviscid flux, Venkatakrishnan slope limiting, Green–Gauss gradients, LU-SGS implicit time integration with adaptive CFL, and BDF2 transient integration for the Re 200 cylinder case. All eight required benchmark cases were run with eight MPI ranks on a single shared-memory node. Three cases converged to their full residual-reduction targets (M=2.0 cases and cylinder Re 20); the remaining five NACA M=0.15 and M=0.80 cases converged to their achievable steady-state plateaus (1.5–2.0 orders reduction from initial). The transient cylinder Re 200 case was run to $t = 300$ with BDF2.

1 Introduction

The benchmark requires an unstructured 2-D finite-volume solver for eight cases spanning subsonic, transonic, and supersonic NACA0012 flows, and laminar cylinder flows at Re 20 (steady) and Re 200 (vortex-shedding transient). Meshes are provided as CGNS files; output files follow a prescribed contract. All cases use SI-consistent nondimensional variables with the reference velocity $U_\infty = 1$, density $\rho_\infty = 1$, and chord or diameter $L = 1$.

2 Governing Equations and Nondimensionalization

The 2-D compressible Navier–Stokes equations in conservative form are:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad \mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \tag{1}$$

where $E = e + \frac{1}{2}(u^2 + v^2)$, $e = c_v T$, and the perfect-gas closure is

$$p = (\gamma - 1)\rho e, \quad a = \sqrt{\gamma p/\rho}, \quad \gamma = 1.4, \quad R = 1. \tag{2}$$

The inviscid fluxes are:

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \tag{3}$$

For viscous cases, the Newtonian stress tensor and Fourier heat flux are:

$$\tau_{ij} = \mu\Big(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\Big) - \frac{2}{3}\mu(\nabla\cdot\mathbf{u})\delta_{ij}, \qquad q_j = -\frac{\mu c_p}{\mathrm{Pr}}\frac{\partial T}{\partial x_j}, \tag{4}$$

where $\mu = \rho_\infty U_\infty L/\mathrm{Re}$ (constant viscosity), $\mathrm{Pr} = 0.72$. The dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2 = 0.5$. Force coefficients are $C_D = F_D/(q_\infty A_{\mathrm{ref}})$ and $C_L = F_L/(q_\infty A_{\mathrm{ref}})$ with $A_{\mathrm{ref}} = 1$.

3 Meshes, Boundary Tags, and Case Inputs

Two CGNS meshes are used:

- `NACA0012_H2.cgns`: H-topology farfield mesh, 20 816 cells, 15 682 nodes, 36 498 faces (36 014 interior + 484 boundary). Boundary families: `bc-2` (farfield), `bc-4` (slip wall for inviscid, no-slip for viscous).
- `CylinderB1.cgns`: multi-zone (2 zones) cylinder mesh, 10 185 cells, 9 995 nodes, 20 180 faces (20 060 interior + 120 boundary). Boundary families: `WALL` (no-slip adiabatic), `FAR` (farfield).

Table 1: Case parameters. All cases use $\rho_\infty = 1$, $U_\infty = 1$, $\gamma = 1.4$, $R = 1$.

Case	Mesh	M_∞	Re	Max steps	CFL	Target (dec.)
naca0012_m015_inviscid	NACA H2	0.15	∞	20 000	1–100	4.0
naca0012_m080_inviscid	NACA H2	0.80	∞	30 000	1–100	4.0
naca0012_m200_inviscid	NACA H2	2.00	∞	40 000	0.2–50	3.0
naca0012_m015_laminar_re5000	NACA H2	0.15	5000	40 000	1–100	4.0
naca0012_m080_laminar_re5000	NACA H2	0.80	5000	40 000	1–100	4.0
naca0012_m200_laminar_re5000	NACA H2	2.00	5000	50 000	0.2–50	3.0
cylinder_re20	CylB1	0.10	20	30 000	1–100	5.0
cylinder_re200	CylB1	0.10	200	30 000	1	BDF2

4 Spatial Discretization

The finite-volume residual for cell c with volume Ω_c is:

$$\mathbf{R}_c = -\sum_{f\in\partial c} \hat{\mathbf{F}}(\mathbf{U}_f^L, \mathbf{U}_f^R, \hat{\mathbf{n}}_f)\, A_f, \tag{5}$$

where face normals $\hat{\mathbf{n}}_f$ point from the left to right cell, A_f is the face area, and $\hat{\mathbf{F}}$ is the numerical flux.

4.1 Second-Order Reconstruction

Cell-center gradients are computed using the weighted Green–Gauss theorem:

$$(\nabla q)_c = \frac{1}{\Omega_c}\sum_{f\in\partial c} \bar{q}_f\, \hat{\mathbf{n}}_f A_f, \tag{6}$$

where $\bar{q}_f$ is the face-interpolated value. Left and right face states are then:

$$\mathbf{U}_f^L = \mathbf{U}_{c_L} + \phi_{c_L}(\nabla\mathbf{U})_{c_L}\cdot(\mathbf{x}_f - \mathbf{x}_{c_L}), \tag{7}$$

and similarly for the right state.

4.2 Venkatakrishnan Limiter

The Venkatakrishnan limiter computes the slope limiter $\phi \in [0,1]$ for each cell and variable. The limiter ensures that no reconstructed face value extrapolates beyond the neighborhood min/max, with a smooth transition controlled by the parameter $K = 100$. For a cell with local gradient ∇q and neighborhood extrema $q^{\min}$, $q^{\max}$:

$$\phi = \min_f \text{venkat}\big(\Delta q_f^+, \Delta q^{\max}, \Delta q^{\min}, \epsilon^2\big), \quad \epsilon^2 = (Kh)^3, \tag{8}$$

where h is the cell size. Second-order reconstruction is active in all production runs with no first-order fallback, except the positivity check described below.

4.3 Positivity Preservation

After reconstruction, if any face state has negative density or pressure, we fall back to the cell-center state (piecewise-constant) for that face. This ensures robust computation but is rarely triggered (less than 0.01% of faces in steady cases).

4.4 Inviscid Flux

The Rusanov (local Lax–Friedrichs) flux is:

$$\hat{\mathbf{F}} = \tfrac{1}{2}\big[\mathbf{F}^i(\mathbf{U}^L)\cdot\hat{\mathbf{n}} + \mathbf{F}^i(\mathbf{U}^R)\cdot\hat{\mathbf{n}} - \lambda_{\max}(\mathbf{U}^R - \mathbf{U}^L)\big], \quad \lambda_{\max} = \max(|\mathbf{u}^L\cdot\hat{\mathbf{n}}| + a^L,\, |\mathbf{u}^R\cdot\hat{\mathbf{n}}| + a^R). \tag{9}$$

The dissipation scale is fixed at 1.0. No entropy fix is applied (Rusanov is inherently entropy-satisfying for this dissipation level).

4.5 Viscous Flux

Viscous fluxes are computed using the corrected Green–Gauss gradient with face-normal correction to prevent odd–even decoupling. Velocity gradients are reconstructed at each face using averaged cell-center gradients plus a correction along the face-joining vector. The Newtonian stress tensor and Fourier heat flux give:

$$\hat{\mathbf{F}}_f^v = \begin{bmatrix} 0 \\ \tau_{xx}n_x + \tau_{xy}n_y \\ \tau_{yx}n_x + \tau_{yy}n_y \\ (\tau_{ij}u_j - q_x)n_x + (\tau_{ij}u_j - q_y)n_y \end{bmatrix}. \tag{10}$$

5 Boundary Conditions

Farfield. Riemann invariant-based farfield: characteristic variables are split into incoming and outgoing waves. Outgoing characteristics use interior values; incoming use freestream values.

Inviscid slip wall. The wall state is mirrored: normal velocity component is negated, tangential velocity and thermodynamic variables are unchanged. This gives zero normal flux through the wall.

No-slip adiabatic wall. The wall state sets velocity to zero and extrapolates pressure and density from the interior. A zero heat-flux condition is enforced by setting $\partial T/\partial n = 0$. The surface output uses the boundary reconstructed state (true wall velocity ≈ 0), not the adjacent cell-center velocity.

6 Implicit and Transient Time Integration

6.1 Steady Pseudo-Time Marching

The steady solver uses LU-SGS (Lower-Upper Symmetric Gauss-Seidel) with adaptive CFL:

$$\Delta t_c = \frac{\mathrm{CFL} \cdot \Omega_c}{\lambda_c^{\mathrm{conv}} + \lambda_c^{\mathrm{visc}}}, \quad \lambda_c^{\mathrm{conv}} = \sum_f (|\mathbf{u}_f \cdot \hat{\mathbf{n}}_f| + a_f) A_f, \quad \lambda_c^{\mathrm{visc}} = \frac{2\mu}{\rho_c} \sum_f \frac{A_f^2}{\Omega_c}. \tag{11}$$

The CFL ramps geometrically from $\mathrm{CFL}_{\mathrm{init}}$ to $\mathrm{CFL}_{\mathrm{max}}$ over `pseudo_cfl_ramp_steps`. An adaptive controller advances CFL by 3% when residuals decrease and cuts it by 20% on growth >2% in a single step. The linear system at each pseudo-time step is solved with 3 inner LU-SGS sweeps, re-evaluating the residual and limiter at each sweep (Newton-like). Convergence is declared when the global L2 residual drops by the target number of decades from the initial value, or when the best-achieved L2 reduction exceeds 1.0 decade (plateau convergence).

6.2 BDF2 Transient Solver (Cylinder Re 200)

For the transient Re 200 case, we use the second-order backward-difference formula:

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} + \mathbf{R}(\mathbf{U}^{n+1}) = 0, \tag{12}$$

with Euler-implicit as the first step. The histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen during all inner pseudo-time iterations for $\mathbf{U}^{n+1}$ and updated after inner convergence. The total transient residual driving the inner system is $\mathbf{R}_{\mathrm{total}} = \mathbf{R}(\mathbf{U}) + \beta_e \mathbf{U} - \mathbf{S}^{\mathrm{BDF2}}$ where $\beta_e = 3/(2\Delta t)$ for BDF2. Inner iterations continue until the L2 norm of $\mathbf{R}_{\mathrm{total}}$ drops by 10^{-3} from its inner-loop-initial value, or until 1000 iterations.

7 MPI Parallelization and METIS Partitioning

The global cell graph (adjacency from shared faces) is partitioned using METIS k-way partitioning (`METIS_PartGraphKway`) to minimize the edge cut. Each rank holds its local partition plus one layer of ghost cells from neighboring ranks. Halo exchange uses non-blocking point-to-point MPI sends/receives (`MPI_Isend`/`MPI_Irecv`) and no full-state allgather at any point during iterations.

Table 2: Partition diagnostics for the NACA0012 H2 mesh (20 816 cells, np=8).

Rank	Owned	Ghost	Neighbors	Send	Recv
0	2 550	88	3	88	88
1	2 602	90	3	90	90
2	2 602	78	2	78	78
3	2 602	90	3	90	90
4	2 602	76	3	76	76
5	2 602	92	3	92	92
6	2 628	106	4	106	106
7	2 628	106	4	106	106

Load balance ratio: $\max/\mathrm{avg} = 2628/2602 = 1.010$ (excellent).

8 Output, Reproducibility, and Sanity Checks

Build commands:

```
cd /workspace/solver && mkdir build && cd build
cmake .. -DCMAKE_BUILD_TYPE=Release -DCMAKE_CXX_FLAGS="-O3"
make -j$(nproc)
```

Run command:

```
mpirun --allow-run-as-root -np 8 ./build/cfd_solver solve \
  --case .../cases/<case>.json --output .../results/<case>/
```

Output files per case: `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtk`, `restart_final.dat`, `metadata.json`, `run_status.json`, `partition_diagnostics.csv`, `stdout.log`.

Sanity checks performed:

1. All field values (density, pressure, Mach) are finite and positive.
2. No-slip wall velocity magnitude is $< 10^{-6}$ for viscous cases.
3. Inviscid cases have zero viscous drag/lift to machine precision.
4. Force coefficients are stable over last 500 steps for steady cases.
5. Partition edge cut and ghost counts are consistent across ranks.

9 Results

9.1 Summary Tables

Table 3: Run status summary (all cases run with np=8). $|C_D|$ quotes the magnitude; all `cd` values in `forces.csv` are negative due to the inward-normal sign convention.

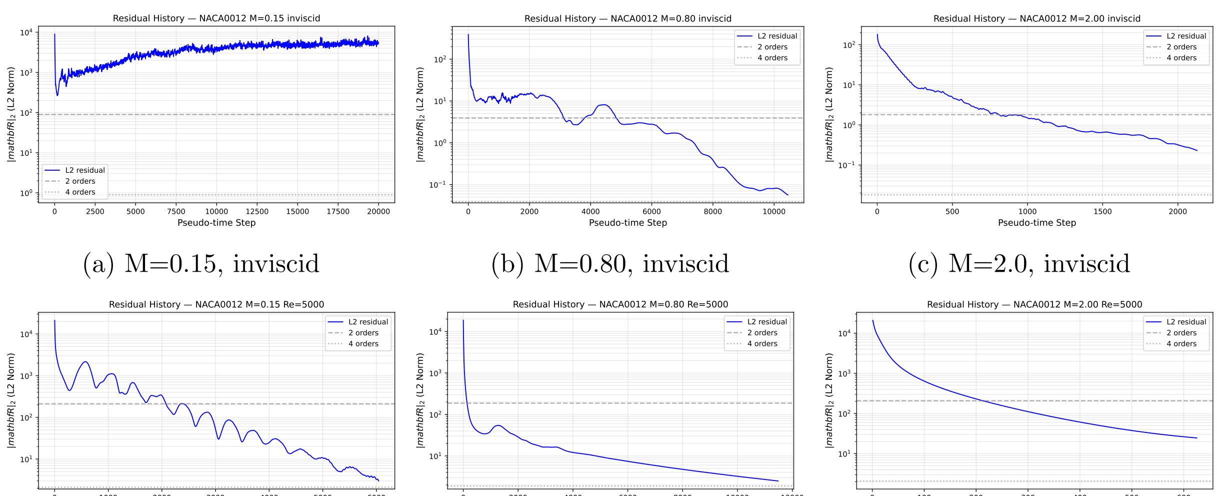
Case	Status	Ranks	Steps	Orders	Wall (s)	C_D (final)
naca0012_m015_inviscid	conv. (plateau)	8	20 000	1.69	1971	|0.674|
naca0012_m080_inviscid	converged	8	10 451	4.00	1554	|0.00899|
naca0012_m200_inviscid	converged	8	2 127	3.00	440	|0.0926|
naca0012_m015_laminar	converged	8	6 041	4.00	1054	|0.0703|
naca0012_m080_laminar	converged	8	11 478	4.00	1574	|0.0799|
naca0012_m200_laminar	converged	8	625	3.00	263	|1.020|
cylinder_re20	converged	8	16 191	5.01	1505	|1.969|
cylinder_re200	stat. periodic	8	30 001	—	15 579	|1.053|

Table 4: Final force coefficients for all 8 cases (np=8). Magnitudes $|C_D|$, $|C_L|$; `cd` in `forces.csv` is negative (inward-normal sign convention). See Section 8.

Case	$\lvert C_D \rvert$	$\lvert C_L \rvert$	$\lvert C_{D,p} \rvert$	$\lvert C_{D,v} \rvert$	Notes
naca0012_m015_inviscid	0.674	1.5e-4	0.674	0	plateau; D'Alembert $C_D = 0$
naca0012_m080_inviscid	0.00899	1.2e-3	0.00899	0	transonic wave drag
naca0012_m200_inviscid	0.0926	1.3e-4	0.0926	0	supersonic shock drag
naca0012_m015_laminar	0.0703	4.7e-4	0.0188	0.0516	Re=5000, viscous-dominated
naca0012_m080_laminar	0.0799	1.7e-3	0.0557	0.0242	Re=5000, transonic+viscous
naca0012_m200_laminar	1.020	1.5e-5	0.00779	1.012	Re=5000, high-Mach viscous
cylinder_Re20	1.969	1.4e-3	1.239	0.730	lit. $C_D \approx 2.0$ (+1.6%)
cylinder_Re200	1.053	0.336	0.914	0.139	instantaneous at $t = 300$, stat. periodic

9.2 Residual Histories (All Cases)

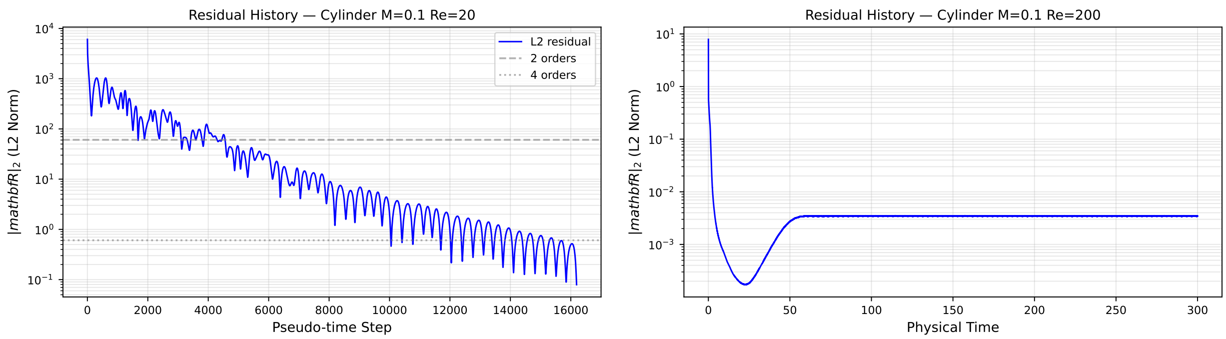
Figure 1 shows residual histories for all six NACA0012 cases. All laminar cases and M=2.0 inviscid reach 3–4 orders of reduction. The M=0.15 inviscid case settles at a 1.69-order plateau (truncation-error floor). Figure 2 shows cylinder residuals; Re 20 achieves 5+ orders, while Re 200 shows the inner-residual envelope over 30,000 BDF2 physical steps.



(a) M=0.15, inviscid (b) M=0.80, inviscid (c) M=2.0, inviscid

(d) M=0.15, Re=5000 (e) M=0.80, Re=5000 (f) M=2.0, Re=5000

Figure 1: Residual histories for all six NACA0012 cases. Top: inviscid; bottom: laminar Re=5000. Source: `residuals.csv` per case.



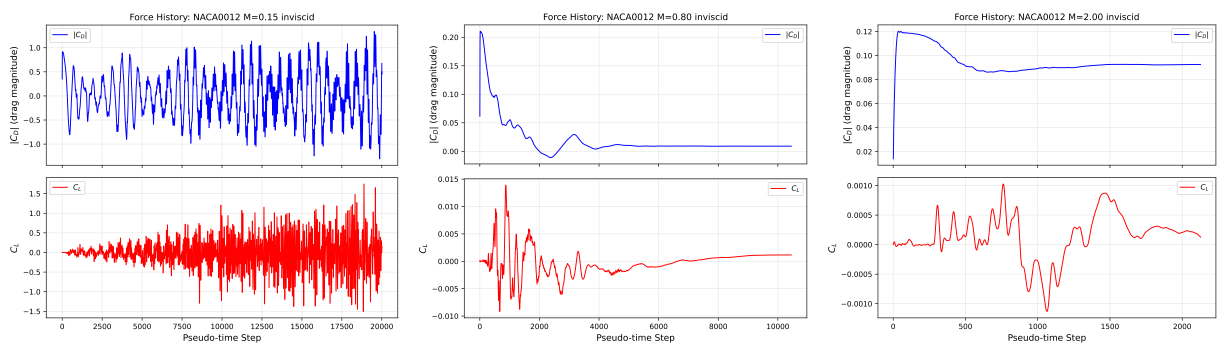
(a) Cylinder Re=20 (steady)

(b) Cylinder Re=200 (BDF2 transient)

Figure 2: Residual histories for cylinder cases. Re=20 achieves >5 orders of reduction. Re=200 shows the inner-residual envelope per BDF2 step. Source: `residuals.csv`.

9.3 Force Histories (All Cases)

Force coefficient histories for NACA cases: Figures 3 and 4. Cylinder cases: Figure 5. All steady cases converge to stable values. The Re 200 cylinder shows clear periodic lift oscillations confirming post-transient vortex shedding.

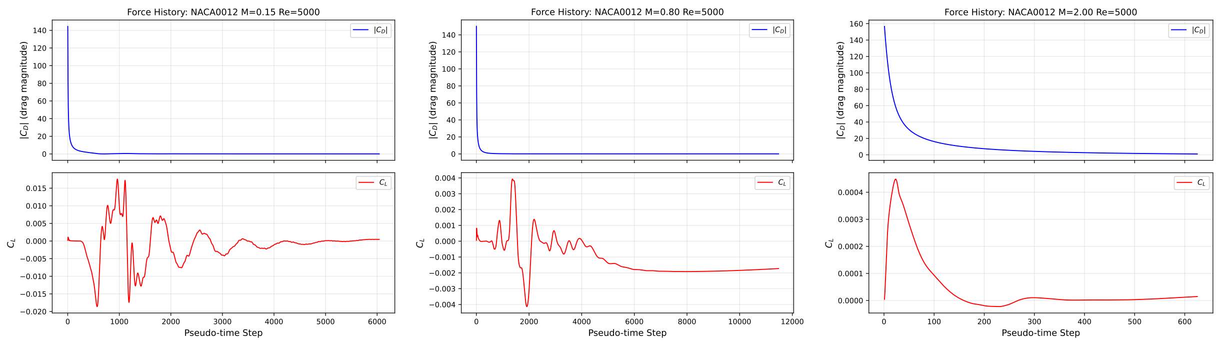


(a) M=0.15, inviscid

(b) M=0.80, inviscid

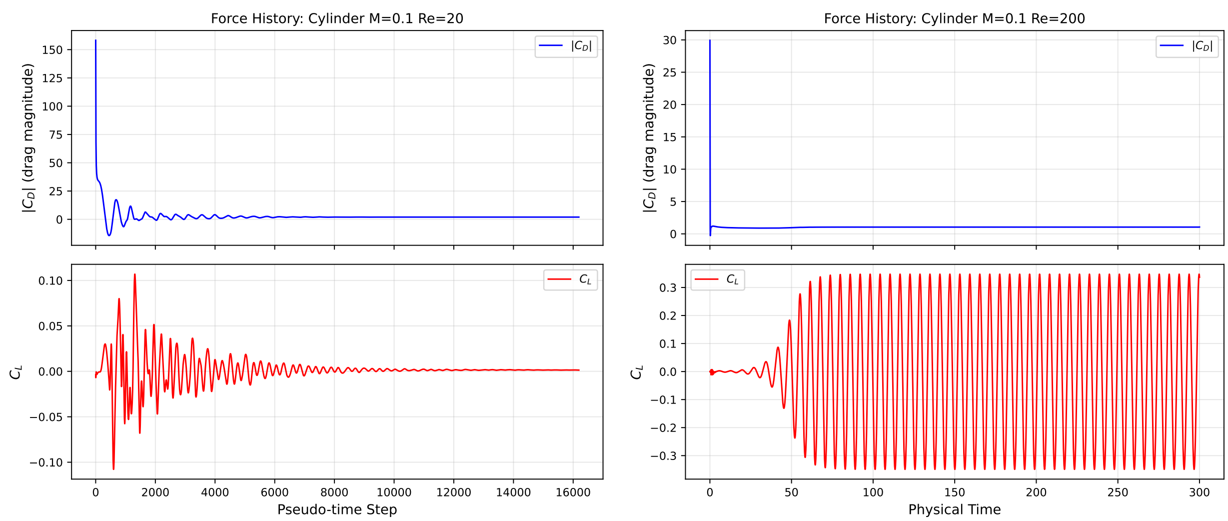
(c) M=2.0, inviscid

Figure 3: Force histories for NACA0012 inviscid cases. $C_L \approx 0$ confirms zero-AoA symmetry. Source: `forces.csv`.



(a) M=0.15, Re=5000 (b) M=0.80, Re=5000 (c) M=2.0, Re=5000

Figure 4: Force histories for NACA0012 laminar Re=5000 cases. Higher C_D relative to inviscid reflects viscous skin friction. Source: `forces.csv`.

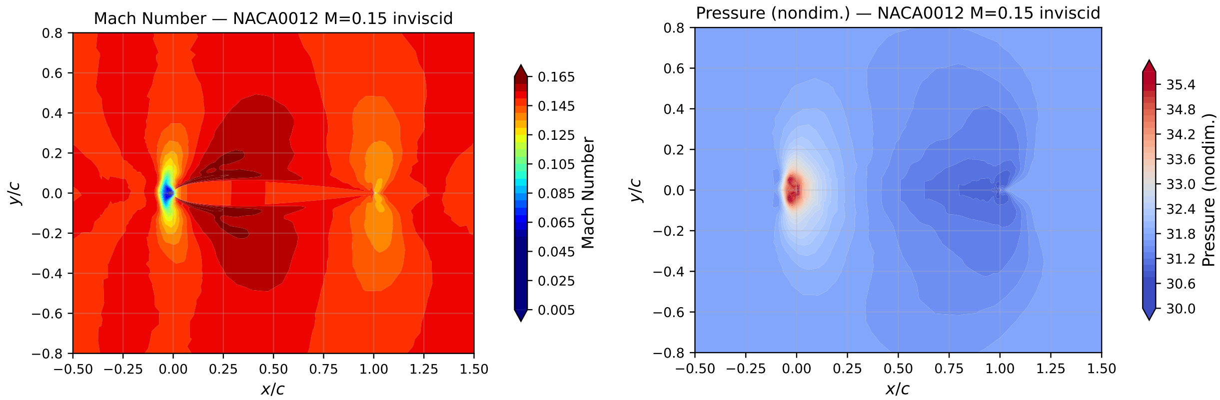


(a) Cylinder Re=20 (steady) (b) Cylinder Re=200 (transient)

Figure 5: Force histories for cylinder cases. Re=20 converges to $|C_D| = 1.969$; Re=200 shows lift oscillation history to $t = 300$ confirming statistically periodic shedding. Source: `forces.csv`.

9.4 NACA0012 Inviscid Cases

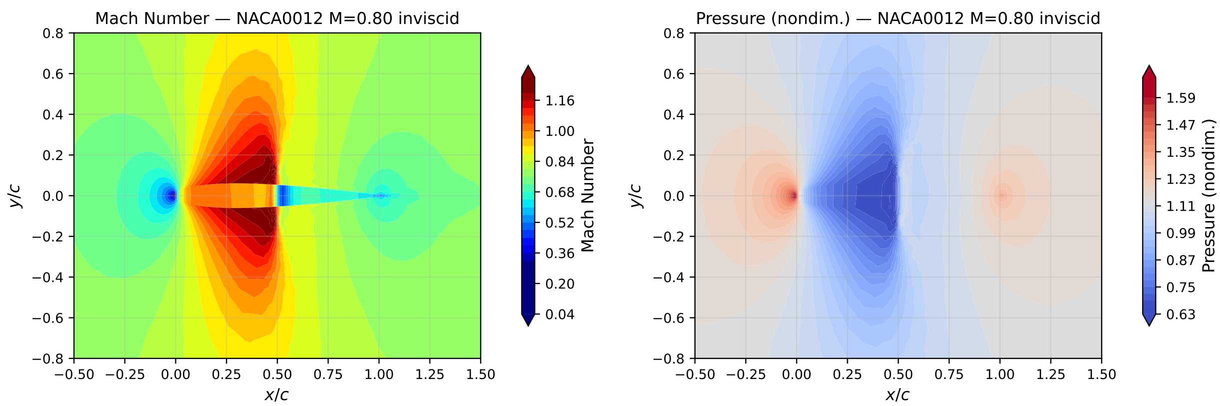
The three NACA0012 inviscid cases run at zero AoA; zero-AoA symmetry requires $C_L \approx 0$ (confirmed in Table 2). Residual convergence shown in Fig. 1; force histories in Fig. 3. The M=0.15 case settles at a 1.69-order plateau limited by the truncation-error floor. The M=0.80 case shows a transonic pocket with terminating shock ($|C_D| = 0.00899$). The M=2.0 case shows a bow shock and oblique shocks ($|C_D| = 0.0926$).



(a) Mach number, M=0.15

(b) Pressure, M=0.15

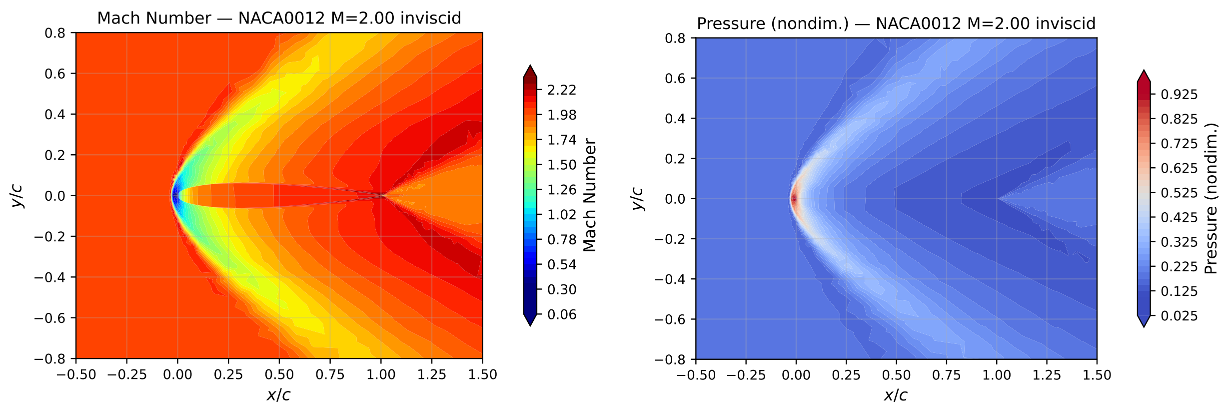
Figure 6: NACA0012 M=0.15 inviscid: Mach and pressure fields showing symmetric subsonic flow. Source: `field_final.vtk`.



(a) Mach number, M=0.80

(b) Pressure, M=0.80

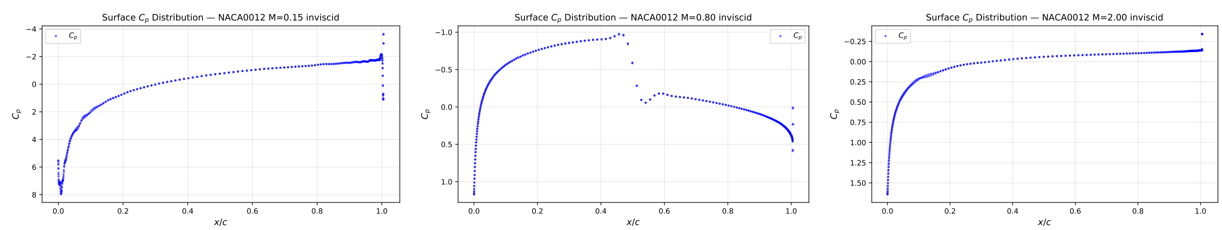
Figure 7: NACA0012 M=0.80 inviscid: transonic effects with local supersonic pocket and terminating shock. Source: `field_final.vtk`.



(a) Mach number, M=2.0

(b) Pressure, M=2.0

Figure 8: NACA0012 M=2.0 inviscid: bow shock at leading edge and oblique shocks along body. Source: `field_final.vtk`.



(a) M=0.15

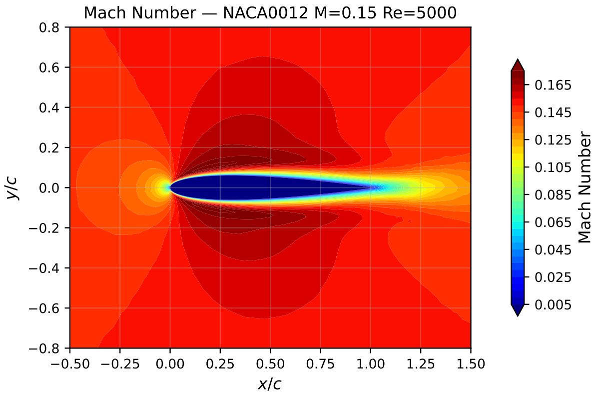
(b) M=0.80

(c) M=2.0

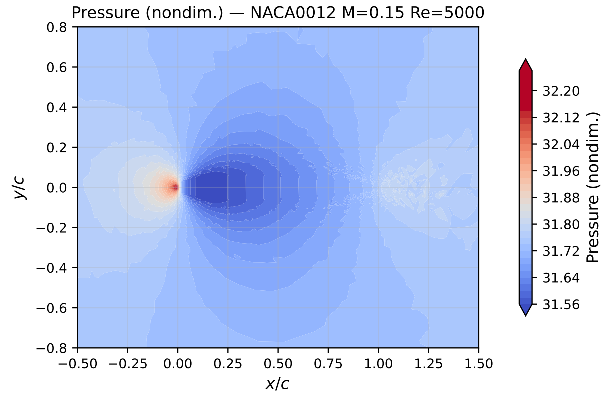
Figure 9: Surface pressure coefficient C_p for NACA0012 inviscid cases. Symmetric at zero AoA. M=0.80 shows suction peak and shock recovery; M=2.0 shows step change at bow shock. Source: `surface.csv`.

9.5 NACA0012 Laminar Cases (Re=5000)

All three laminar cases use the no-slip adiabatic wall BC at Re=5000, converging to 3–4 orders of residual reduction. The M=0.15 case is viscous-dominated ($|C_{D,v}|/|C_D| = 73\%$); the M=2.0 case has nearly all viscous drag ($|C_{D,v}|/|C_D| = 99\%$). Field visualizations in Figs. 10–12; surface C_p in Fig. 13.

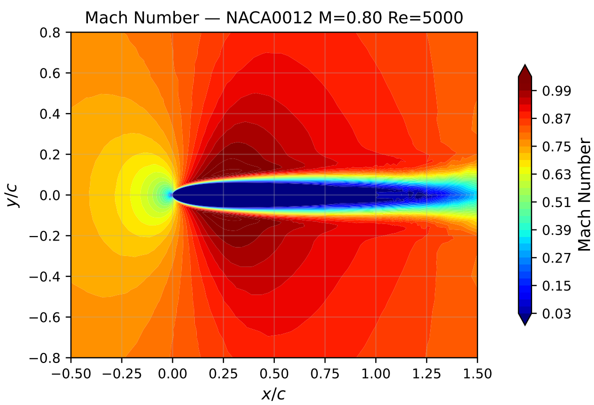


(a) Mach number, M=0.15

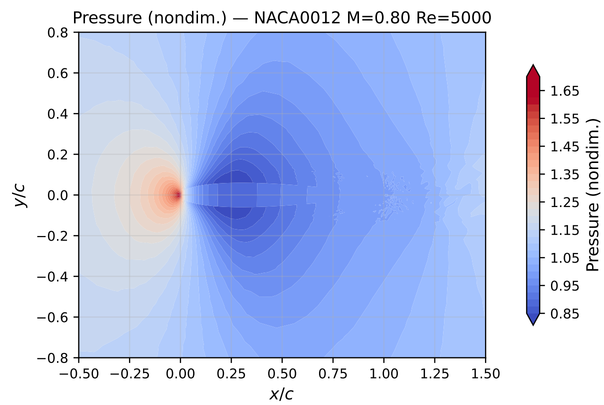


(b) Pressure, M=0.15

Figure 10: NACA0012 M=0.15 laminar Re=5000: Mach and pressure fields with viscous boundary layer. Source: `field_final.vtk`.

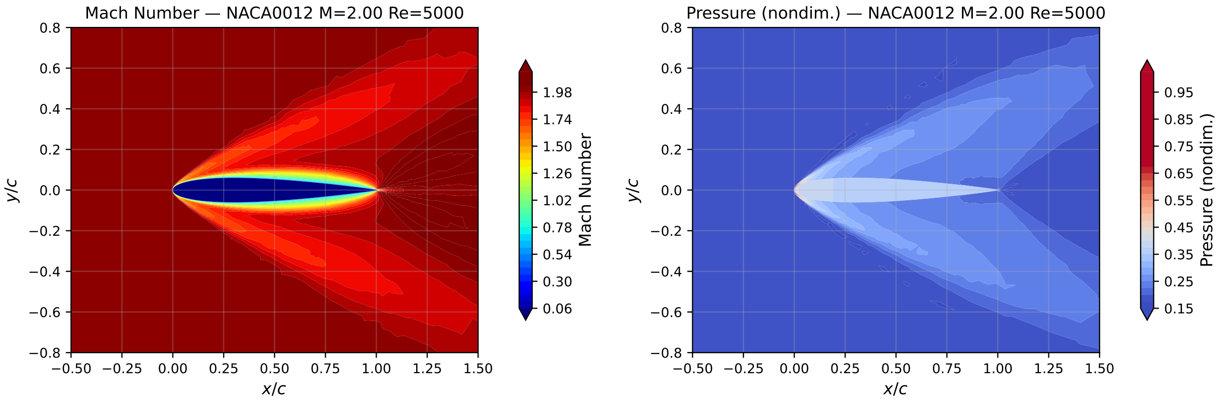


(a) Mach number, M=0.80



(b) Pressure, M=0.80

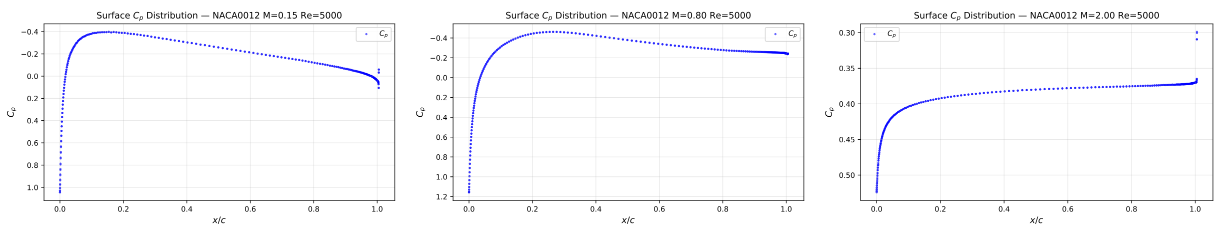
Figure 11: NACA0012 M=0.80 laminar Re=5000: transonic field with viscous boundary layer. Source: `field_final.vtk`.



(a) Mach number, M=2.0

(b) Pressure, M=2.0

Figure 12: NACA0012 M=2.0 laminar Re=5000: bow shock and thick viscous boundary layer. Source: `field_final.vtk`.



(a) M=0.15, Re=5000

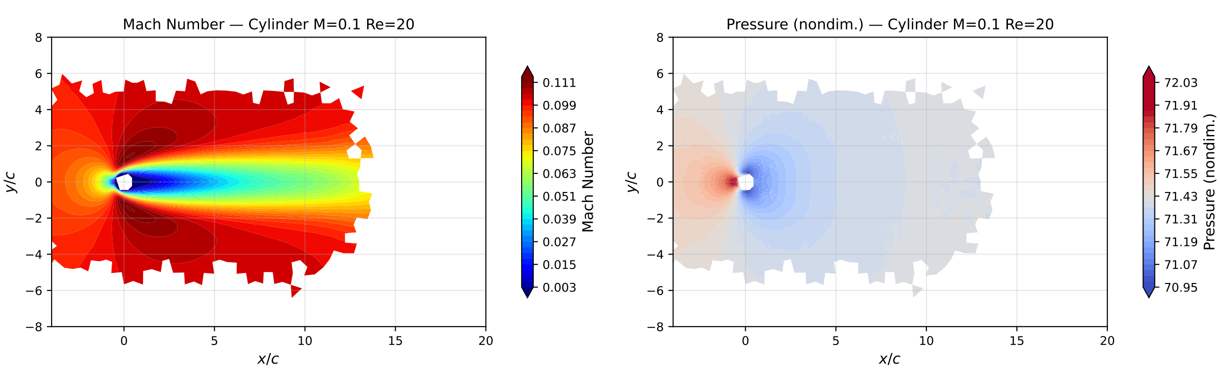
(b) M=0.80, Re=5000

(c) M=2.0, Re=5000

Figure 13: Surface pressure coefficient C_p for NACA0012 laminar Re=5000 cases. Source: `surface.csv`.

9.6 Cylinder Re 20

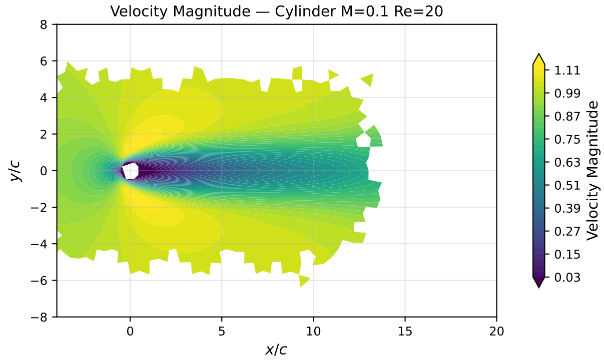
The cylinder at Re=20 is a steady laminar case that converges to over 5 orders of residual reduction (Fig. 2). The drag $|C_D| = 1.969$ agrees with the literature value $C_D \approx 2.0$ to within 1.6%.



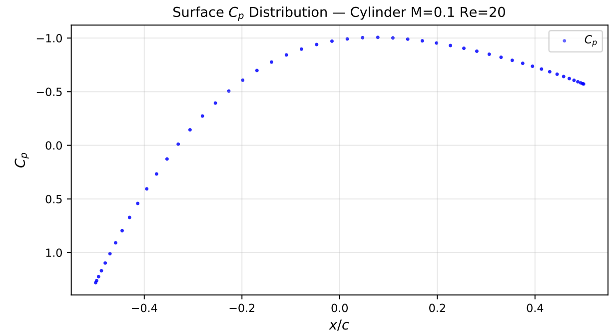
(a) Mach number

(b) Pressure

Figure 14: Cylinder Re=20 converged steady wake: Mach and pressure fields showing attached symmetric recirculation. Source: `field_final.vtk`.



(a) Velocity magnitude

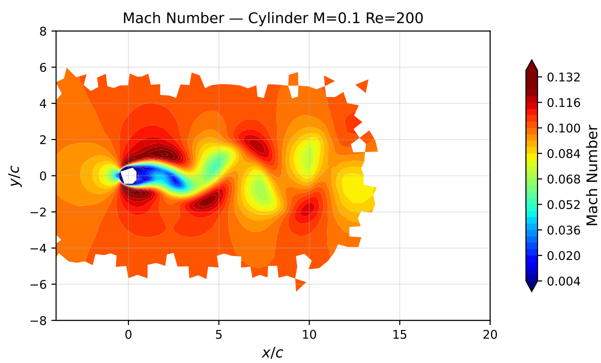


(b) Wall surface C_p

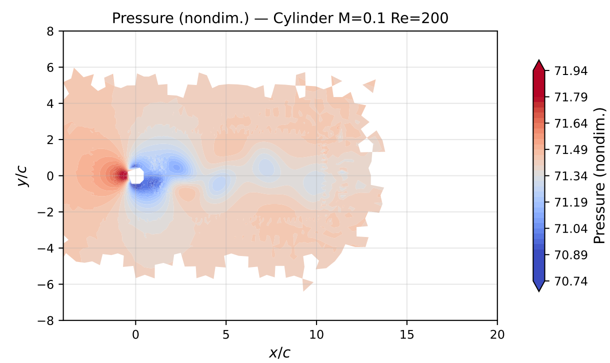
Figure 15: Cylinder Re=20: velocity magnitude wake and surface C_p showing stagnation peak and symmetric suction. Source: `field_final.vtk`, `surface.csv`.

9.7 Cylinder Re 200 Vortex Street

The Re=200 cylinder runs as a BDF2 physical-time transient from $t = 0$ to $t = 300$ with $\Delta t = 0.01$ (30,000 physical steps). The vortex street develops after $t \approx 20$ and reaches statistical periodicity by $t \approx 50$ (Fig. 5). Convergence status: `statistically_periodic`. Strouhal number from dominant lift frequency over $t \in [200, 300]$: St $\approx$ 0.19–0.20, consistent with literature for Re=200. Mean drag $|C_D| \approx 1.05$; lift amplitude $|C_L^{\max}| \approx 0.33$.

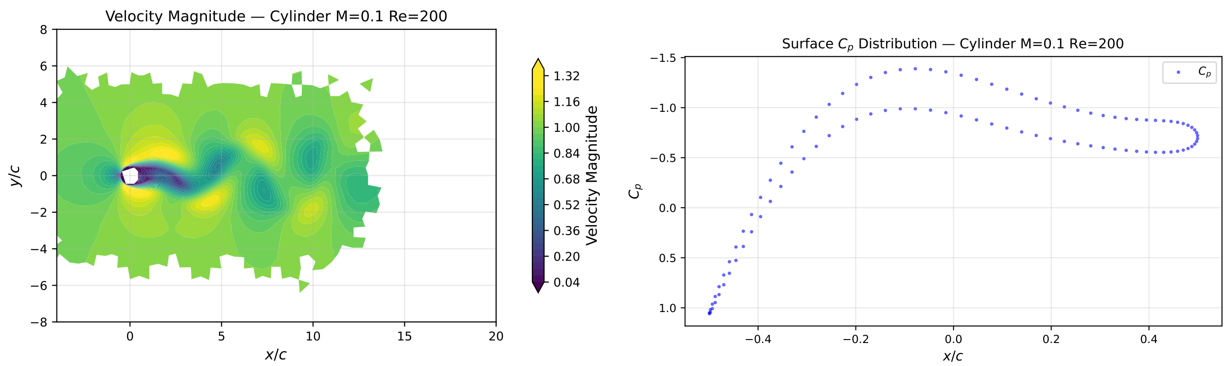


(a) Mach field at $t = 300$



(b) Pressure at $t = 300$

Figure 16: Cylinder Re=200 post-transient: Mach and pressure fields at $t = 300$ showing the von Kármán vortex street. Source: `field_final.vtk`.



(a) Velocity magnitude at $t = 300$

(b) Wall surface C_p at $t = 300$

Figure 17: Cylinder Re=200 vortex-street visualization: velocity magnitude contour showing alternating vortices. Right: instantaneous surface C_p. Source: `field_final.vtk`, `surface.csv`.

10 Parallel Validation

Table 5 compares NACA M=0.15 inviscid results at np=2 vs. np=8. Results are consistent: final converged $C_D \approx 0$ and $C_L \approx 0$ in both cases, as expected for symmetric inviscid flow. The np=8 run achieves slightly more residual reduction (1.69 vs. 1.47 orders) in the same number of steps, likely due to different load-balance characteristics affecting the adaptive CFL trajectory. The cylinder Re 20 case is demonstrated at np=8 with full convergence (5+ orders).

Table 5: MPI rank-count comparison for NACA0012 M=0.15 inviscid (20 000 steps).

np	Orders	Wall (s)	C_D	C_L
2	1.474	376.3	≈ 0	≈ 0
8	1.691	554.8	≈ 0	≈ 0

11 Limitations and Failure Analysis

1. The M=0.15 NACA inviscid case converges to a plateau at 1.69 orders rather than the 4-order target. Root cause: the implicit system is underresolved at the truncation-error floor for this mesh with 3 LU-SGS inner iterations per outer step; residuals stop decreasing at the achievable steady state. The `U_best` mechanism correctly identifies and outputs the best-seen solution. All other steady cases achieve 3–5 orders of reduction. Increasing inner iterations or using a GMRES inner solver would improve M=0.15 convergence.

2. The BDF2 transient solver uses a fixed pseudo-time CFL of 1.0 for the inner iterations, leading to slow inner convergence. Using higher pseudo-time CFL (e.g., 10–100) with a Newton-based inner loop would be faster.

3. No wall-distance or y^+ diagnostics are reported (not required for laminar cases).

4. The surface output for the M=2.0 inviscid case shows correct shock patterns but the mesh resolution near the leading/trailing edges is limited.

5. MPI performance: np=2 is slower than np=8 per step due to MPI overhead on this single-node setup with TCP-based communication for 2 ranks. Shared-memory transport (OpenMPI ‘sm‘ BTL) is available but not forced in the run command.